

Introduction to Computational Text Analysis



Digital Integration Teaching Initiative (DITI)

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Workshop Goals

- Read, search, and filter texts in Women Writers Online (WWO) using its reading interface and metadata.
- Utilize the Women Writers Vector Toolkit to see how words are used similarly or differently across corpora.
- Understand citation patterns between authors and texts with intertextual networks.
- Think about archival silences and how text production determines what can be accessed.
- **Handouts and slides:** BITLYLINK HERE

About the Women Writers Project

- A long-term research and publication project focused on building a digital collection of rare and less familiar texts by women. Founded at Brown University in 1988, moved to Northeastern in 2013.
- Publishes Women Writers Online (WWO), which is collection of print English-language texts published between 1526 and 1850.
- Offers additional open-access resources such as Women Writers in Context, Women Writers in Review, Intertextual Networks, and the WWP Lab.

Women Writers Online

Accessing WWO

1. Go to: <https://www.wwp.northeastern.edu/wwo/>
2. Click on “Enter WWO” 
3. Use the following credentials:
 - i. **Username:** philosophy
 - ii. **Password:** wwoDemo!
4. Once you sign in, you will arrive at the WWO interface.

WWO: Key Features

The screenshot shows the Women Writers Online (WWO) website interface. At the top is a navigation bar with links for 'About', 'Help', 'WWP', and 'Feedback'. The main content area is divided into several sections:

- Search across textbase:** A search bar with the placeholder 'Text to find' and a 'Search' button. A red circle with the number '1' is placed over the search bar.
- Filter:** A section with a red circle and the number '2'. It includes filters for 'Genre' (drama: 71, fiction: 84, non-fiction: 229, verse: 153) and 'Date' (16th century: 30, 17th century: 182, 18th century: 141, 19th century: 129).
- Results:** A list of search results. A red circle with the number '3' is placed over the 'Results: 482 texts' header. The results list includes entries such as 'Adams, Hannah — The History of the Jews, 1812', 'Adams, Hannah — A Summary History of New-England, 1799', 'Aikin, Lucy — Epistles on Women, 1810', 'Anger, Jane — Jane Anger Her Protection for Women, 1589', 'Ariadne — She Ventures, and He Wins, 1696', 'Askew, Anne — The First Examination of Anne Askew, 1546', 'Askew, Anne — The Latter Examination of Anne Askew, 1547', 'Askew, Anne — The Two Examinations of Anne Askew (Foxe), 1563', 'Astell, Mary — Reflections Upon Marriage, 1706', 'Astell, Mary — A Serious Proposal to the Ladies, 1694', 'Augustine, Sister Magdalen — Introductory Material to The History of the Angelical Virgin Glorious S. Clare, 1635', and 'Bacon, Ann (Cooke), trans. (from John Jewel) — An Apology or Answer in Defence of the Church of England, 1564'.
- Timeline view:** A vertical timeline on the right side of the page. A red circle with the number '4' is placed over the timeline. The timeline shows a vertical axis with years (1526, 1550, 1600, 1650) and a series of red dots representing the lives of the writers.

1 Search the collection

2 Filter (genre/century)

3 Results

4 Timeline view

Exploring WWO: Search Within One Text

- In WWO, open Cavendish's *Philosophical Letters* (listed as Cavendish, Margaret (Lucas), Duchess of Newcastle — **Philosophical Letters**, 1664)
- Toggle between the reading view and the original typography view. What does each one show you?
- Use the "**Search Document**" feature (top left) and search for the word "philosophy." How many times does the word occur in *Philosophical Letters*?
- Click a result to read the passage. How is Cavendish using the word "philosophy"?

Exploring WWO: Search Across the Collection

- Return to the main WWO interface and rerun “philosophy” in the **search box on the top left**.
- How many texts mention the word “philosophy” across the collection?
- Look at the timeline feature on **the right side of the screen**. When does it appear the most?
- Narrow the results by genre and date: try **Non-Fiction** → **Essays**, then narrow to **17th century**. How many texts are left?
- Search for “natural philosophy” (include the quotation marks for an exact match). How many texts contain that exact phrase? Whose text has it the most?

Women Writers Vector Toolkit

Overview of the Vector Toolkit

- A model “reads” a corpus and learns which words are likely to appear in similar contexts.
- Words that are close are used in similar ways in the texts, even though they are not necessarily synonyms.
- The higher the similarity score, the closer the relationship.
- Each model reflects only the corpus it was trained on.

Query term:

November

	Word	Similarity to word(s)
1	november	1
2	october	0.890950880656073
3	april	0.882562541064641
4	january	0.875357193362418
5	june	0.873807841829722
6	february	0.872507475338337
7	december	0.869709107890752
8	september	0.859504885122682
9	july	0.853978078069506
10	august	0.837725489689188

Accessing WWVT

1. Go to: <https://wwp.northeastern.edu/lab/wwvt/>
2. Click on “Click to explore the Word Vector Interface.”

Click to explore the Word Vector Interface

3. Once you sign in, you will arrive at the WWVT interface.

WWVT: Key Features

The screenshot shows the Women Writers Vector Toolkit (WWVT) interface. On the left sidebar, there is a 'Model' dropdown menu (1) set to 'WVO Full Corpus' and a 'Number of Words' slider (2) set to 10. The main area has tabs for 'Home', 'Compare', 'Clusters', 'Operations', and 'Visualization' (3). The 'Home' tab is active, displaying a welcome message and a section titled 'WVO Full Corpus' with a description of the corpus. Below this is a 'Query term:' input box (4). To the right of the input box, there is a table (5) showing results. The table has two columns: 'Word' and 'Similarity to word(s)'. The results are as follows:

	Word	Similarity to word(s)
1		
2	the	
3	and	
4	of	

1 Choose a model

2 Number of words

3 Additional Tools

4 Query term

4 Results

Explore the Vector Toolkit (1/2)

- Choose a model and type the word “philosophy” in the text box (no quotation marks needed). Raise the Number of Words to 30. You can click a word to see it in WWO.
- In the **Compare** tab, choose WWO and VEP Early Modern models and enter the same word. What changes do you notice?
- In the **Clusters** tab, hit Reset until you find a cluster that interests you. What is connecting the words in this particular cluster?

Explore the Vector Toolkit (2/2)

- In the **Operations** tab, combine two words such as "natural" and "philosophy." What do the results suggest?
- In the **Visualization** tab, plot words that are similar in nature, such as "physician" and "surgeon." Which words are gathering around each one?

Women Writers Intertextual Networks

Accessing WWIN

1. Go to: <https://wwp-test.northeastern.edu/intertextual-networks>
2. Use the following log-in details:
 - i. **Username:** philosophy
 - ii. **Password:** ww0Demo!
3. Once you sign in, you will arrive at the WWIN interface.

Note: This interface can be slow! Don't worry if it takes a minute to load.

WWIN: Key Features

Women Writers: Intertextual Networks

Explore ▾ About ▾ Research & Teaching ▾

What did pre-Victorian women writers read? How did they use and cite their reading, and how did their writing shape the literate cultures of their times? What can parody, adaptation, quotation, and paraphrase tell us about the web of knowledge in which these texts participate?

Explore Intertextual Networks and find out!

[Bibliography](#)

3

The books, articles, poems, songs, and other works referenced in Women Writers Online texts.

[Intertextual Gestures](#)

4

The quotations, citations, and other references made by women writers in the WWO collection.

[Authors in WWO](#)

1

The women writers of WWO, drawing from and engaging with other works in their own creations.

[Topics & Genres](#)

2

"Current events," "philosophy," and the many other kinds of writing represented in the bibliography.

[Elizabeth Ogilvy Benger](#) in [The Female Geniad](#), 1791:

[Miss Seward](#) unites so many exquisite beauties in her poetry, that it would be difficult to pronounce where she excels the most. In 1779 she composed the [Prize Monody on the death of Mr. Garrick](#); in 1780 she publish'd an [Elegy on Captain Cooke](#); with a Hymn to the Sun; the [unhappy fate of Major André](#) received the same tribute. She has since written [Louisa](#), her Poetical Novel (a new species of composition) and beside those already mentioned, several miscellaneous pieces.

WWIN: Authors in WWO

Sort method

Name



Direction

Ascending: 0-9, A-Z



Apply

Jump to section

1

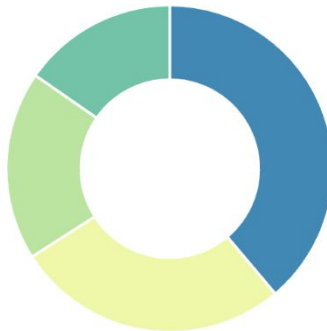
Astell, Mary

Astell is the author of 8 [works in the bibliography](#), of which 7 [works are referenced in Women Writers Online](#).

2

There are 59 [intertextual gestures](#) in Astell's 2 [works published in WWO](#):

3



4

1

Browse Authors

2

Author's Works

3

Intertextual Gestures

4

Gesture Mix

WWIN: Topics and Genres

Sort method

Name



Direction

Ascending: 0-9, A-Z



Apply

Jump to section

1

Natural philosophy

2

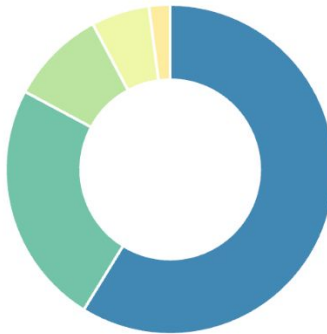
Works on the study of nature.

There are 13 [works tagged “Natural philosophy”](#) in the bibliography.

3

There are 54 [references to “Natural philosophy” works](#):

4



1

Browse topics

2

Topic

3

Works tagged

4

References

WWIN: Bibliography

4,626 results

Page 1 of 93

Previous

Next

Page

1

Go

Filter results

Contributor

A person (or organization) whose efforts shaped the work.

58 [Barbauld, Anna Laetitia \(Aikin\)](#)

43 [Dryden, John](#)

41 [Cavendish, Margaret \(Lucas\), Duchess of Newcastle](#)

41 [Horace \(Q. Horatius Flaccus\)](#)

38 [Pope, Alexander](#)

36 [Shakespeare, William](#)

29 [Haywood, Eliza \(Fowler\)](#)

Sort method

Number of References to

Direction

Descending: Z-A, 9-0

Apply

Number of References to	Title	Contributors	Publication Date	Publication Location	Topics & Genres	Links
953	Psalms	[unknown]			Sacred text	 
933	Matthew	[unknown]			Sacred text	 
912	Referenced work not found	[unknown]	not applicable	not applicable		 
812	Revelation	[unknown]			Sacred text	 
803	Isaiah	[unknown]			Sacred text	 

1

Sort by reference

2

Reference count

3

Filters (contributor, genre, etc)

4

Topics & Genres



Reading an Intertextual Gesture

Source Text [?] 1 		Gesture [?] 2		Referenced Work [?] 3 	
Text	Topics & Genres [?]	Text of the Gesture	Gesture Type [?]	Text	Topics & Genres [?]
Adams, Hannah. <i>A Summary History of New-England</i>. 1799.    	History	“Hail, O hail My much lov’d native land! New Albion hail! The happiest realm, that, round his circling course, The all searching sun beholds. With read more	quote	Dwight, Timothy. <i>Greenfield Hill: A Poem in Several Parts</i>. 1794.  	Poetry

Cavendish Intertextual Gestures (1/2)

Filter results

Gesture Type

- 462 [quote](#)
- 189 [citation](#)
- 83 [title](#)
- 30 [adaptation](#)

Referenced Work

A work which has been utilized in an intertextual gesture within a source text.

- 72 [Cavendish, Margaret \(Lucas\), Duchess of Newcastle. *Philosophical Letters*. 1664.](#)
- 71 [van Helmont, Franciscus Mercurius. *Oriatrike; or, Physick Refined*. 1662.](#)
- 63 [More, Henry. *The Immortality of the Soul*. 1713.](#)
- 61 [No external source.](#)
- 57 [Hobbes, Thomas. *Elements of Philosophy*. 1658.](#)
- 47 [Hobbes, Thomas. *Leviathan*. 1651.](#)
- 35 [Helmont, Jean Baptiste van. "Of the Birth, or Original of Forms." *Van Helmont's Works, Containing His Most Excellent Philosophy, Physick, Chirurgery, Anatomy*. 1664.](#)
- 34 [More, Henry. *Antidote Against Atheism*. 1655.](#)
- 30 [Cavendish, Margaret \(Lucas\), Duchess of Newcastle. *Philosophical and Physical Opinions*. 1655.](#)
- 30 [Descartes, René. *Meditations on First Philosophy*. 1641.](#)

Referenced Work's Topic/Genre

- 319 [Philosophy](#)
- 197 [Medicine](#)
- 193 [Essay](#)
- 106 [Scientific writing](#)
- 72 [Letter](#)
- 42 [Natural philosophy](#)
- 42 [Theology](#)
- 29 [Compendium](#)
- 12 [Sacred text](#)
- 10 [Gender commentary](#)

Cavendish Intertextual Gestures (2/2)

Source Text ⓘ		Gesture ⓘ		Referenced Work ⓘ	
Cavendish, Margaret (Lucas), Duchess of Newcastle. <i>Philosophical Letters</i> . 1664.    	Philosophy. Letter	“Let it be light; Let there be a Firmament in the midst of the Waters, and let the Waters under the Heaven be gathered together, read more	quote	Genesis.  	Sacred text
		“things Present onely have a being in Nature, things Past onely a being in the Memory, but things to come have no being at all;”	quote	Hobbes, Thomas. <i>Leviathan</i> . 1651.  	Philosophy.

Explore Intertextual Networks

- Click on **Intertextual Gestures**
- In the **Source Text** filter, choose Cavendish, Margaret (Lucas), Duchess of Newcastle. *Philosophical Letters*. 1664. How many gestures do you see?
- Check the **Gesture Type**. Which type of gesture is the highest? Which type of gesture is the lowest?
- Check **Referenced Work**. Whose work is Cavendish drawing on the most? Click one to narrow the results.
- Read the **source text** → **gesture** → **referenced work** for any of the results.

Discuss: Archival Silences

Digital collections inherit the gaps of the archives they come from. Women's writing was often judged as not worth keeping, so what is recovered shapes what the search or model can show.

1. The male-authored comparison corpora are far larger than WWO. What does that difference tell us?
2. Which writers, genres, and kinds of texts were archived and later digitized, and by whom?
3. How might one influential author shape the searches, models, and citation patterns we saw today?
4. What can absences of certain words or topics tell us?

For Further Exploration

Data Privacy

- It's important to pay attention to data privacy when using digital resources
- At its simplest, **data privacy** is a person's ability to control what of their personal information is shared and with whom.
- To help you make informed decisions about interacting with digital tools in ways that honor your boundaries with your data and/or personal information, The DITI has prepared a handout on [Data Privacy](#)

Resources

- DITI handouts on [building a corpus](#) and more [links and resources](#) for text analysis
- DITI handout on [troubleshooting text analysis](#)
- NULab [list of resources for text analysis](#)
- [Programming Historian tutorials](#)
- [“Data-Sitters’ Club” tutorials](#)
- Library subject guides on text mining and analysis: [guide on getting started](#), [guide on vendor policies](#)

Thank you!

—Developed by Mel Williams and Sarah Connell

- For more information on DITI, please see: <https://bit.ly/diti-about>
- Schedule an appointment with us! <https://bit.ly/diti-meeting>
- If you have any questions, contact us at: nulab.info@gmail.com